Tectonophysics, 79 (1981) 201—223
Elsevier Scientific Publishing Company, Amsterdam — Printed in The Netherlands

LATE PALEOZOIC ROTATIONS OF CORSICA AND SARDINIA: NEW EVIDENCE FROM PALEOMAGNETIC AND K-Ar STUDIES

J.B. EDEL, R. MONTIGNY and R. THUIZAT

Laboratoire de Paléomagnétisme, Institut de Physique du Globe, 5 Rue Descartes, 67084 Strasbourg Cedex (France)

(Received December 16, 1980; revised version accepted March 13, 1981)

ABSTRACT

Edel, J.B., Montigny, R. and Thuizat, R., 1981. Late Paleozoic rotations of Corsica and Sardinia: new evidence from paleomagnetic and K-Ar studies. Tectonophysics, 79: 201—223.

A detailed paleomagnetic investigation of so-called "Permian" volcanics was undertaken in southeastern and northwestern Sardinia. K-Ar dating on separate minerals representing 30 sites on Corsica and Sardinia showed that the first calc-alkaline cycle had taken place in Westphalo-Stephanian times and the second alkaline one at the end of the Stephanian. 246 paleomagnetic specimens from 28 sites of southeastern Sardinia and 2 sites from the northwest were demagnetized by alternating field and thermal processes. The directions of characteristic magnetizations are $D = 85^\circ$, $I = -6^\circ$, $\alpha_{95} = 5^\circ$ for the volcanics of southeastern Sardinia and $D = 126^\circ$, $I = -1^\circ$ for the Nurra samples. These mean directions differ from those of northwestern Corsica and northern Sardinia (Gallura). The paleomagnetic results display good grouping for each volcanic complex but exhibit a geographical dispersion of the mean directions. This scattering suggests significant motions in Late Hercynian times.

INTRODUCTION

Upper Paleozoic volcanics of calc-alkaline and alkaline nature outcrop extensively in Corsica, Sardinia, southwest France and in northern Italy (Vellutini, 1977). In terms of plate tectonics, the presence of calc-alkaline volcanics implies the existence of active continental margins. One may, therefore, expect large plate or microplate motions similar to those of the Tertiary (Bellon et al., 1977; Edel, 1980).

Several paleomagnetic studies had been carried out on these upper Paleozoic volcanics in Corsica and Sardinia. The investigations in Corsica led some authors (Nairn and Westphal, 1968; Westphal, 1976; Westphal et al., 1976) to deduce an anticlockwise rotation of 30° relative to stable Europe in post-Permian times, while others questioned this conclusion (Storetvedt and Petersen, 1976). Similar studies in northern Sardinia (Zijderveld et al., 1970; Westphal et al., 1976; Storetvedt and Markhüs, 1978) suggested an anticlock-

0040-1951/81/0000—0000/$02.50 © Elsevier Scientific Publishing Company

wise rotation of Sardinia relative to Europe of 70° to 45° after emplacement of the volcanics. The proposed reconstruction suffered from lack of data in northwest and southeast Sardinia and from the high uncertainty in relative and absolute ages of the volcanics. Geochronological investigations on peralkaline granites associated with basalts and alkaline rhyolites of Corsica yielded contradictory results (Bonin et al., 1972; Maluski, 1974, 1977). In Sardinia several K-Ar studies carried out on the Upper Paleozoic volcanics gave ages ranging from 180 to 260 Ma (Cozzupoli et al., 1971; Lombardi et al., 1974).

The aim of this investigation was to correlate the paleomagnetic directions with dates in order to study possible Late Hercynian motions. Paleomagnetic work and K-Ar dating were carried out on the volcanic complexes of Barbagia, Ogliastra, Gerrei and Nurra (Fig. 1). Additional K-Ar measurements were also made on Corsican samples that had already been studied paleomagnetically (Westphal, 1976; Westphal et al., 1976). In this way, the paleomagnetic directions regarding the Upper Paleozoic of both islands might be fruitfully compared.

GEOLOGICAL SETTING

Corsica

The complex of Scandola-Cinto (northwest Corsica) is composed of two different volcanic cycles (Vellutini 1973, 1977; Westphal et al., 1976). The older one, of calc-alkaline type, is composed of trachy-andesites and rhyolitic ignimbrites which overlie metamorphosed rocks and sedimentary layers including coals with Stephanian flora. The younger one is represented by basalts, alkaline and peralkaline rhyolites associated with alkaline granites in ring complexes. Radiometric dates on those ring complexes are contradictory. An age of 232 Ma was obtained by the Rb-Sr method (Bonin et al., 1972) while ^{39}Ar-^{40}Ar investigations (Maluski, 1974, 1977) along with U-Pb ones (Lancelot, in Maluski, 1977) yielded Late Hercynian dates ranging from 280 to 310 Ma.

Sardinia

Upper Paleozoic volcanics outcrop in the north (Gallura), the northwest (Nurra), the southeast (Barbagia, Ogliatra, Gerrei) and the southwest of the island (Fig. 1). In Gallura, Vellutini (1977), Westphal et al. (1976), recognized both volcanic cycles defined in Corsica. The stratigraphy of the other massifs is ill-defined. Attempts were made to assess the age of volcanism. On paleontological grounds, an Autunian age was attributed to the sedimentary basins of Barbagia and Gerrei (Novarese, 1917; Lauro et al., 1963; Pecorini, 1974). Nevertheless, the flora might also be of Upper Carboniferous age (J. Doubinger, pers. comm., 1980). Use of the conventional K-Ar method on

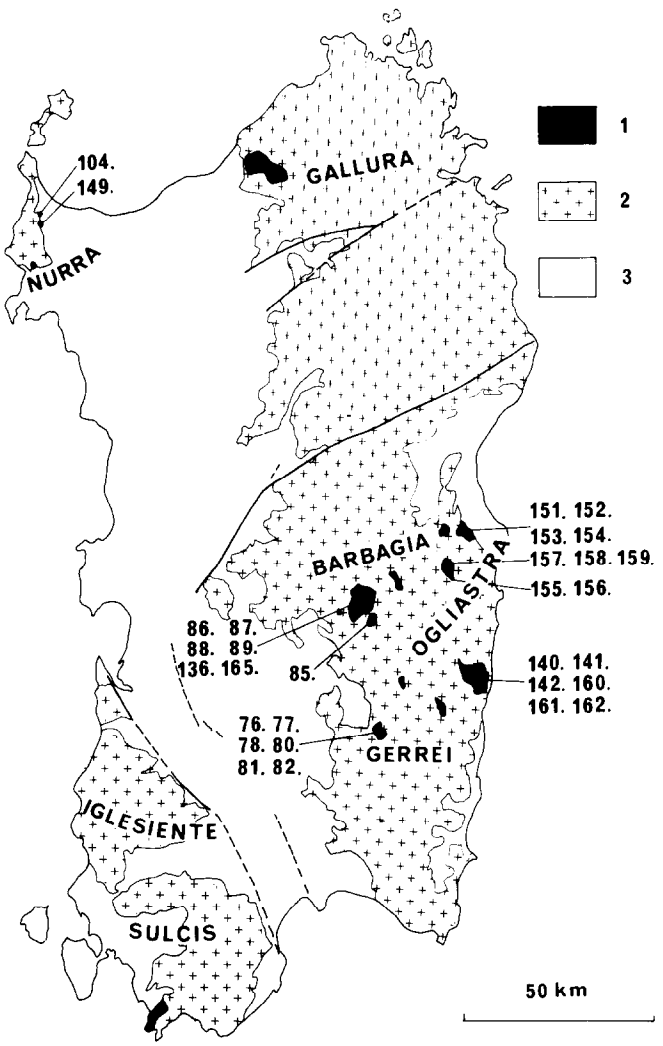


Fig. 1. Location of sampling sites. *1* = Upper Paleozoic volcanics; *2* = Paleozoic basement; *3* = Post-Paleozoic outcrops.

whole rocks from Seui-Seulo (Barbagia) (Cozzupoli et al., 1971) yielded ages in the range of 250—265 Ma. Subsequent determinations on whole rocks from southeast and southwest Sardinia gave ages of 197—260 Ma (Lombardi et al., 1974). The same authors obtained ages of 204—197 Ma respectively on pyroclastites and ignimbrites from Nurra. In the light of these results, volcanism would have been spread through the Permian and Lower Trias.

PALEOMAGNETIC STUDY

Sampling

The upper Paleozoic basin of Seui-Seulo in Barbagia (Fig. 1) is characterized by alternating layers of continent sediments and pyroclastic volcanics overlying a corrugated metamorphosed Silurian basement (Lauro et al., 1963). The age of those formations might be lower Autunian from the flora (Novarese, 1917; Lauro et al., 1963) as well as from radiometric K-Ar dates (Cozzupoli et al., 1971). Six sites of ignimbrites (86, 87, 88, 136), one site of andesite (85) and two sites of ignimbritic tufs (89, 165) have been sampled for analysis.

The volcanic complex of Ogliastra, situated east of Barbagia, exhibits three main outcrops, all with a NNW -SSE trend. The series of Baunei comprises black, red and grey ignimbrites overlying conglomeratic sediments which are in unconformity with the basement. Lombardi et al. (1974) obtained an age of 250 Ma on an ignimbrite which corresponds to our site 151. Black (site 151) and red (sites 153, 154) ignimbrites have been sampled, as well as tufs (site 152). Similar rocks have also been found in the series of Talana -Villagrande Strisaili. Chemical compositions of the ignimbrites are identical to those of the Baunei series. K-Ar ages of 235 Ma and 228 Ma have been measured on a breccia and a rhyolitic ignimbrite respectively (Lombardi et al., 1974).

Three sites (157, 158, 159) correspond to the black ignimbrites lying at the bottom of the series and two sites represent porphyritic dikes (155, 156) which intrude the metamorphic basement.

The massif of Monte Ferru di Tertenia lies to the south along the Tyrrhenian coast (Fig. 1). The ignimbrites which compose the major part of the volcanic complex, display similar compositions to those of Baunei and Talana—Villagrande. A K-Ar determination on a rhyolitic ignimbrite (Lombardi et al., 1974) yielded an age of 223 Ma (Lombardi et al., 1974). Ignimbrites (sites 140, 141, 160), andesites (142, 162) and tuffs (site 161) have been sampled.

In Gerrei, the basin of Escalaplano presents the following succession (Pecorini, 1974):

— a lower sedimentary attributed to the Lower Permian.

— a lower volcanic series composed of green to red-brown ignimbrites (sites 80, 81) and of grey to pink tuffs (site 78).

— an upper sedimentary series.

— an upper volcanic andesitic layer (sites 76, 77, 82).

In Nurra, small outcrops of red ignimbritic and pyroclastites occur within clastic sediments of Permo-Triassic age close to the contact with the Paleozoic basement (Pecorini, 1962). Lombardi et al. (1974) have determined two ages of 197 Ma and 204 Ma for these volcanics. Two sites have been sampled (104, 149), located between Monte Giusta and the metamorphic basement.

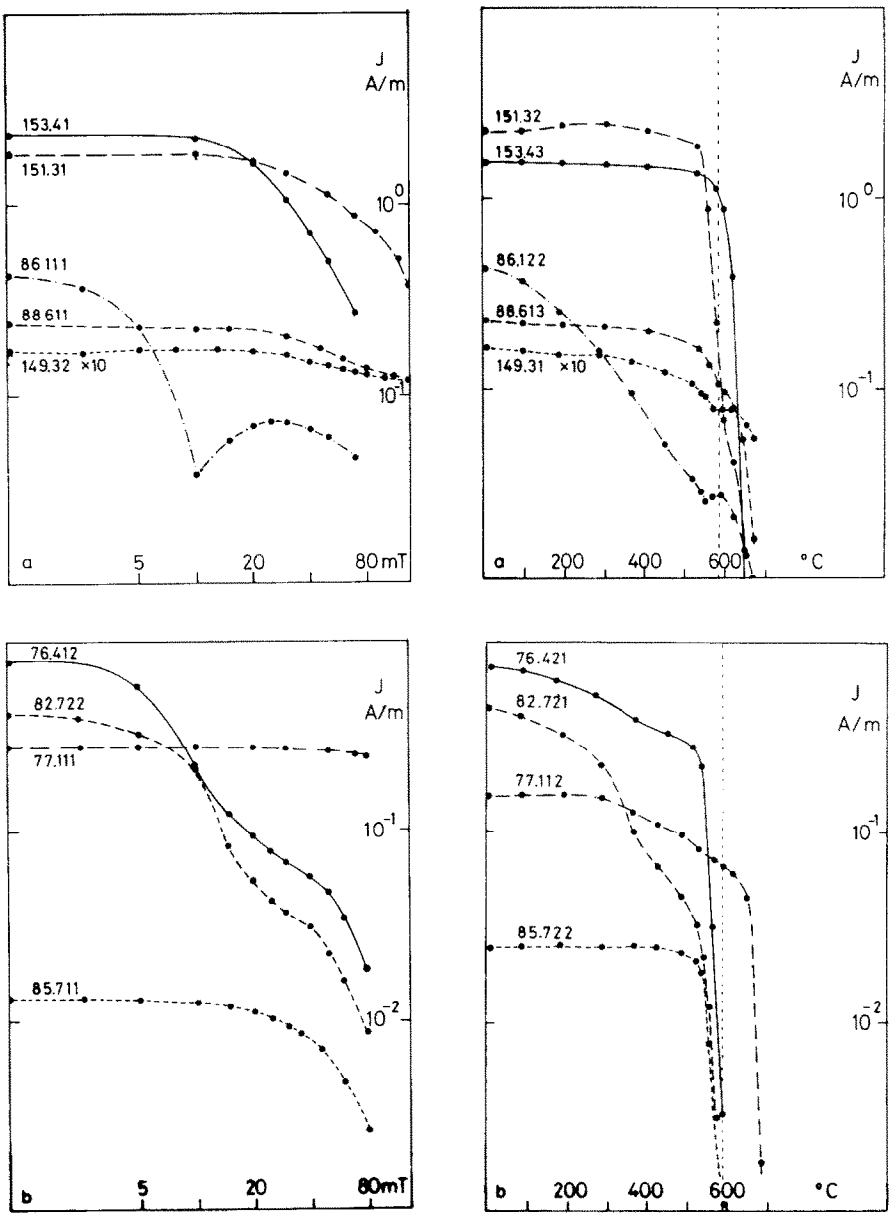


Fig. 2. Typical A.F. and thermal demagnetization curves of ignimbrites (a) and andesites (b).

Altogether, 154 samples, divided into 246 specimens and representing 30 sites of Sardinian Paleozoic volcanics, have been analyzed. Standard paleomagnetic techniques have been applied to these specimens. 13 additional samples have been collected for radiometric dating.

Rock magnetism

The measurements are performed with a Digico magnetometer. Susceptibilities are measured in a field of 0.7 mT at 10 kHz with a susceptibility bridge.

The ignimbrites and the andesites yield intensities of NRM ranging from 10^{-2} to 1 A/m with a maximum of population situated around $2 \cdot 10^{-1}$ A/m. The susceptibilities lie between 10^{-4} and 10^{-1} SI units. The Koenigsberger factors are dispersed around 2. The viscosity factor rarely exceeds 15% of the NRM (Edel, 1980). It is insignificant in most of the ignimbrites.

Alternating field and thermal demagnetizations

The aim of the study is to assess the directions of magnetization acquired by the volcanics when cooling. The samples have, therefore, been demagnetized by an alternating field as well as by thermal processes, in order to eliminate secondary magnetizations.

The high coercivities of the ignimbrites did not in many cases, allow a complete AF demagnetization (Fig. 2a). Alternating fields of only 150 mT could be reached so that a thermal treatment was necessary in many instances.

The andesites generally display blocking temperatures ranging from 550°C to 580°C which are characteristic of titanomagnetites poor in titanium. Intermediate coercivities allow the andesites to be demagnetized to 1/10 of the NRM (Fig. 2b). Site 77 is an exception. It shows a very high coercivity and a blocking temperature of 670°C, both of which are characteristic of haematite.

The results of thermal demagnetization of the ignimbrites allow us to distinguish three types, characterized by the minerals that carry the NRM: magnetite, titanohaematite or both (Fig. 2a). Specimens with a unique blocking temperature of 580°C are rare. In most cases, such as sites 151, 154, 157, 158, 159, 160, magnetite supports more than 90% of the NRM while the residual is due to titanohaematites (specimen 151.32, Fig. 2a). Specimens with only haematite are also rare (specimen 153.43, Fig. 2a). Most of the ignimbrites contain both magnetite and titanohaematite in variable proportions as depicted by Fig. 2a. The occurrence of titanohaematites very often explains the uncommonly high coercivities.

Secondary and primary magnetizations

Three types of secondary magnetization are observed:

(1) Viscous remanent magnetization. This is generally very weak. This weakness increases with the titanohaematitic context of the rock. At 100°C or 5 mT, the viscous remanent magnetization (VRM) completely disappears.

(2) "Weak" secondary magnetization. This appears in sites where mag-

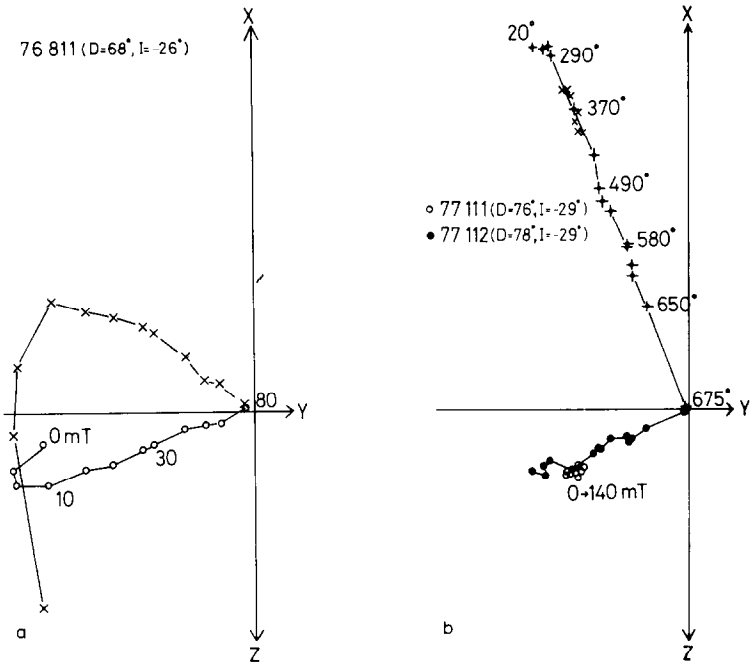


Fig. 3. Typical demagnetization of andesites. (a) dots in *XOY* plan, crosses in *YOZ* plan; (b) dots in *XOY* plan, crosses in *XOZ* plan.

netite dominates, i.e., the "porfiriti" and some ignimbrites. It is characterized by intermediate blocking temperatures of 350°—400°C and by a coercivity of 10—20 mT (Fig. 2b). The demagnetization of those components is generally easy (Fig. 3a).

(3) "Hard" secondary magnetization. This is observed in many specimens which contain magnetite as the carrier of the main magnetization and shows a residual magnetization due to titanohaematites after thermal or A.F. treatment. The absence of a residual implies that the main magnetization vector carried by magnetite converges to the origin (154.31, Fig. 4a); whenever the residual exists, the same vector diverges (154.11, Fig. 4a; 160.13, Fig. 4b). Within a given site, all main vectors corresponding to magnetite are well grouped whereas those carried by titanohaematites are scattered. Then, the converging-circles method yields a secondary direction that is anti-parallel to the one due to magnetite.

All these observations were also made by Storetvedt and Markhüs (1978) on the ignimbrites of Gallura. They showed that the volcanics had been oxidized and remagnetized in a normal field, whereas all primary magnetizations were reversed. Storetvedt and Petersen (1976), on similar formations of Corsica, regarded the residual as primary magnetizations and the important magnetizations carried by magnetite as secondary. This conclusion is, there-

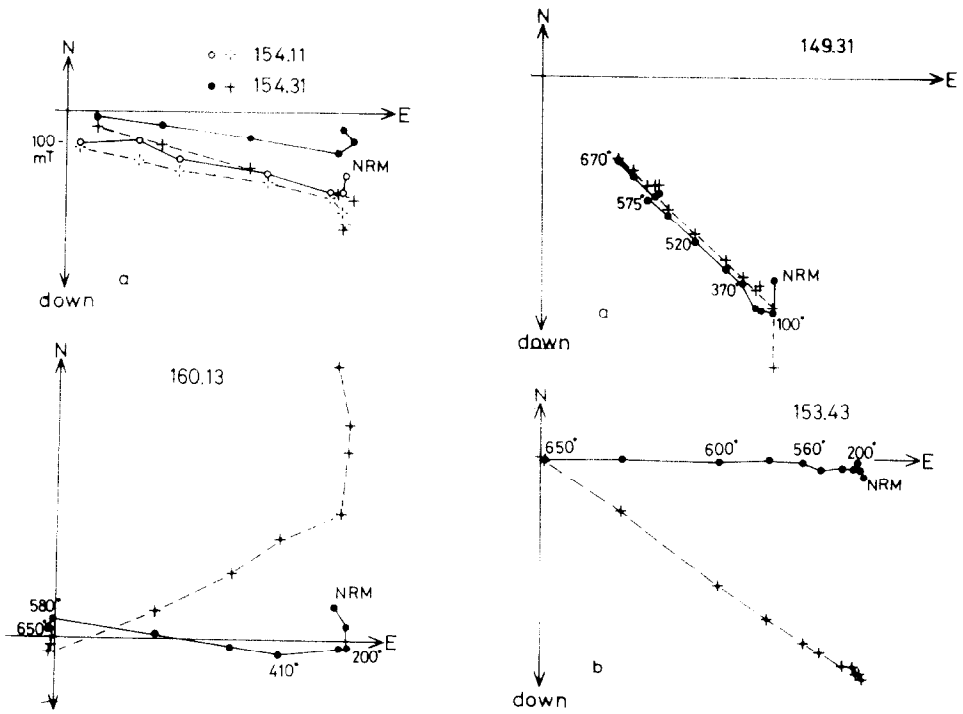


Fig. 4. Typical demagnetizations of ignimbrites with a primary magnetization due to magnetite and a hard secondary magnetization due to haematite (154.11; 160.13).

Fig. 5. Typical demagnetizations of ignimbrites.

fore, contradicted by subsequent observations in Gallura and by ours in south-east Sardinia. Whenever magnetite is the initial magnetic mineral, oxidation can create "hard" secondary magnetizations which still remain after demagnetization.

The characteristic magnetizations of andesites are usually due to magnetite. Site 77 which has been sampled in the same formation as site 76 is an exception (77.112 and 76.421, Fig. 2b). The andesites have been completely oxidized, probably during cooling. This is supported by thermo-magnetic curves and by observations of the isothermal remanent magnetization in progressively increased fields up to 0.9 T. Site 76 (Fig. 4a, magnetite) and site 77 (Fig. 4b, haematite), display identical directions.

The characteristic magnetizations of ignimbrites are carried by magnetite, titanomagnetite or both. This shows that oxidation has occurred during the emplacement of the volcanics. Primary directions are similar, irrespective of the magnetization carrier (Fig. 5a, b).

Characteristic directions (Table I, Fig. 6).

The volcanics of south-east Sardinia form huge masses, so that dip measurements are often difficult or even impossible to make. Nevertheless, in several cases, the occurrence of "flames" allows us to determine dip on the specimens. The directions cluster to the east. The samples from Ogliastra, where numerous dip measurements have been performed, display the best grouping. The dispersion caused by doubtful demagnetizations or lack of dip

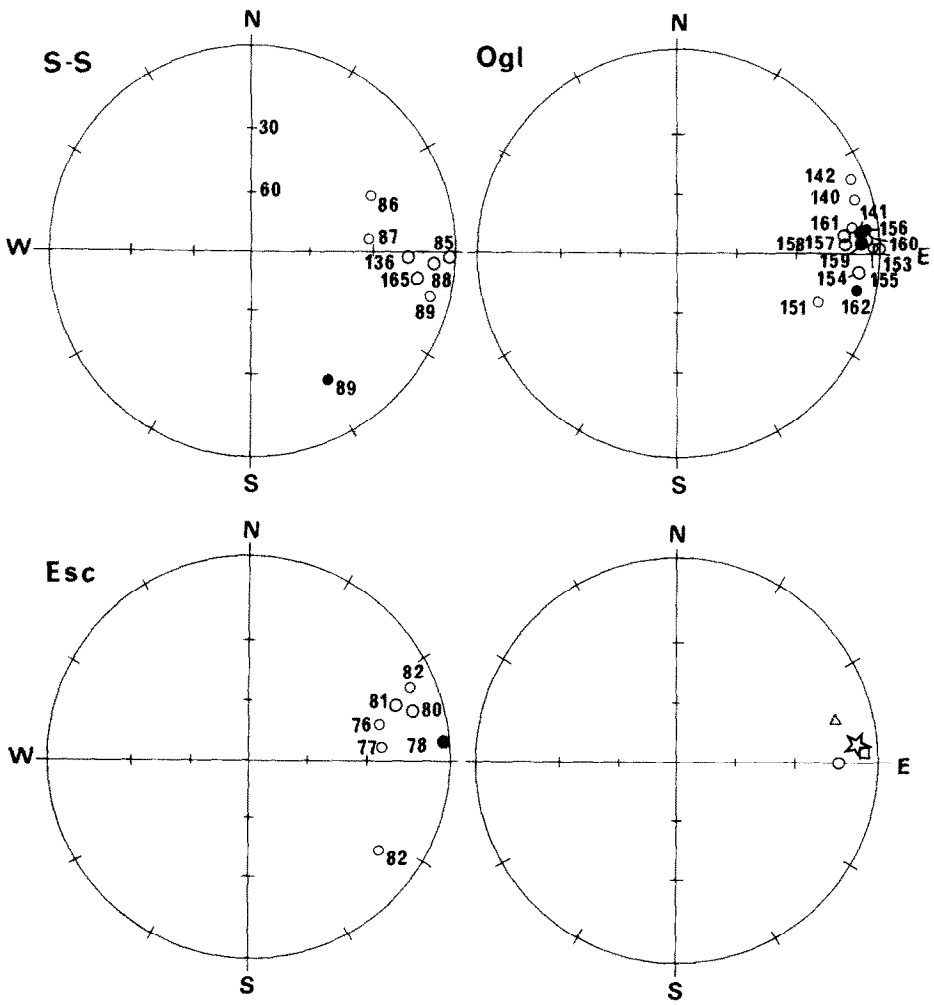


Fig. 6. Stereograms of directions of characteristic magnetizations obtained on sites of Barbagia (*S-S*), Ogliastra (*Ogl*) and Gerrei (*Esc*). Mean directions for Barbagia (○), Ogliastra (□), Gerrei (△) and southeast Sardinia (*). Open symbols = negative inclinations; full symbols = positive inclinations.

TABLE I

Paleomagnetic directions obtained in the Upper Paleozoic volcanics of southeast and northwest Sardinia

Site	Loc.	X	Y	Dip	a	N/N$_o$	µ/n/n$_o$	D	I	R	k	α_{95}	PS
				SEUI-SEULO (BARBAGIA)									
85	Seui	9 18 32	39 51 10	~ 0	1	6/7	11/11/13	91	- 1	10.9352	154	4	a
86	Seui	9 16 56	39 52 52		1	4/4	4/4/4	65	-25	3.9910	334	5	i
87	Seui	9 15 18	39 52 43		1	3/5	4/4/8	83	-31	3.8548	21	21	i
88	Seui	9 15 10	39 52 25	30N85	1	7/7	10/13/13	93	- 6	9.9224	116	4	i
89	Seui	9 15 10	39 52 19			2/4	6/6/6	103	- 6	5.9362	78	8	t
						2/4	4/4/4	149	18	3.9969	997	3	
136	Seui	9 15 14	39 52 35	22N100	1	5/5	6/6/6	92	-16	5.9870	387	3	i
165	Seui	9 15 10	39 52 21	30N40-60	1	4/4	4/4/4	99	-12	3.9763	127	8	t
				BAUNEI (OGLIASTRA)									
151	Baunei	9 36 53	40 03 31	35N230 ?	1	2/4	6/7/12	109	-19	5.9016	51	9	i
152	Baunei	9 36 24	40 04 25	.....		DIVERGENT	.....						t
153	Baunei	9 37 58	40 03 08	35N70	1	5/5	6/6/6	88	0	5.9385	81	7	i
154	Baunei	9 39 15	40 02 38	35N30	1	6/6	7/7/7	96	- 6	6.9730	223	4	i
				TALANA-VILLAGRANDE (OGLIASTRA)									
155	Villagrande	9 32 35	39 57 42		1	3/3	6/6/6	88	- 2	5.9762	211	5	p
156	Villagrande	9 32 11	39 57 32		1	4/4	4/4/5	82	4	3.9548	66	4	p
157	Talana	9 31 38	40 00 32	variable	1	4/4	7/7/9	84	-12	6.9507	122	5	i
158	Talana	9 31 55	40 00 30	30N215	1	3/3	6/7/7	86	-12	5.9276	69	8	i.a?
159	Talana	9 32 11	40 00 29	35N245	2	4/5	12/11/15	87	5	11.8712	85	5	i
					1	4/5	11/11/15	93	33	10.9662	296	3	

				Mt FERRU-TERTENIA (OGLIASTRA)									
140	Tertenia	9 37 27	39 42 32	~0 ?	1	8/8	11/11/11	75	30	10.9242	132	4	i
				35N60 ? 70	1	8/8	11/11/11	73	- 5	10.9242	132	4	i
141	Tertenia	9 36 18	39 44 10	45N330	1	5/5	5/5/5	84	6	4.8101	21	17	i
142	Tertenia	9 40 33	39 42 45	30N200	1	5/5	6/6/8	68	- 4	5.5353	11	21	a
160	Tertenia	9 40 16	39 43 43	35N205	2	5/5	7/7/12	69	-22	6.9615	156	5	i
					1	5/5	8/10/12	86	- 4	7.9832	418	3	
161	Tertenia	9 37 24	39 42 50		1	3/4	3/4/4	81	- 8	2.9368	32	22	t
162	Tertenia	9 37 29	39 42 44		?	1/2	2/2/2	101	6				a
				ESCALAPLANO (GERREI)									
76	Escalaplano	9 20 00	39 35 36	15N330 ?	1	9/9	18/19/21	76	-23	17.5575	38	6	a
77	Escalaplano	9 19 58	39 35 41	15N330 ?	1	4/4	5/5/5	85	-24	4.9925	538	3	a
78	Escalaplano	9 21 01	39 35 41	15N340	1	4/5	4/5/9	85	2	3.9735	113	9	t
80	Escalaplano	9 18 37	39 36 43	~0	1	8/8	9/10/10	73	-11	8.9302	115	5	i
81	Escalaplano	9 19 28	39 37 17	17N220	1	7/7	8/8/8	70	-16	7.9841	442	3	i
82	Escalaplano	9 20 34	39 36 11			4/7	6/7/14	67	- 8	5.8014	25	14	a
						2/7	5/5/14	125	-15	4.9282	56	10	
				NURRA									
104	Canaglia	8 15 00	40 49 10	20N145	1	6/6	6/6/6	128	4	5.9921	637	3	i
149	Canaglia	8 14 23	40 47 45	25-45 N130-145	1	5/5	9/10/10	124	- 6	8.7576	33	9	i

Summary :

South-East Sardinia (Barbagia, Ogliastra, Gerrei) { N= 15 (tectonic correction) D= 85, I = -6, k = 50, α_{95}= 5
{ N= 27 (all sites) D= 85, I = -7, k = 27 α_{95}= 5,5

North-West Sardinia (Nurra) { N= 2 D=126, I = -1

Legend:

Site = site number; *Loc.* = topographic map 1/25.000; *X* = longitude; *Y* = latitude; *Dip* = direction and dip of maximum slope; *a* = primary (1) or secondary (2) magnetization; N/N_0 = number of samples taken for the mean direction computation (N) and total number of samples (N_0); $\mu/n/n_0$ = number of specimens taken for the mean direction computation (μ), number of specimens which show the considered magnetization (n), total number of specimens (n_0); D = declination; I = inclination; R, k, α_{95} = Fisher statistic parameters; *PS* = petrographic and stratigraphic denomination. *a* = andesites; *i* = ignimbrites; *t* = tufs; *p* = porphyries.

correction does not significantly affect the mean direction which is identical whether the 14 optimum sites or all of the 27 sites are taken into account. The mean direction is therefore: $D = 87°$, $I = -6°$, $\alpha_{95} = 5°$ and the corresponding pole position $\lambda = 0°$, $\phi = 284°$ E. It remains similar in all of the

TABLE II

K-Ar dating of the Upper Paleozoic volcanics of Corsica and Sardinia (in Corsica and Gallura the samples have been collected by Westphal, 1976 and Westphal et al., 1976)

Sample			K_2O (%)	$rad.^{40}Ar$ (10^{-11} mol/g)	$\frac{100.rad.^{40}Ar}{Total\ ^{40}Ar}$	Calculated age (± 1σ) in Ma (1) (a)	Calculated age (± 1σ) in Ma (2) (b)
Location and Rock type	Sample Number	Mineral					
OTA (Corsica) Gabbro (2nd cycle)	*138	Hb	0.447	19.41	81	273 ± 9	279 ± 10
	3015	Hb	0.256	11.81	68	289 ± 10	295 ± 10
	3015	Pl	0.295	10.32	69	223 ± 8	228 ± 8
	3016	Hb	0.288	12.23	80	267 ± 9	273 ± 10
	3017	Hb	0.455	20.45	88	282 ± 10	288 ± 10
	3017	Bt	6.540	295.7	97	284 ± 8	290 ± 8
	3017	Pl	0.295	10.45	74	226 ± 8	231 ± 8
	3019	Hb	0.505	20.84	84	260 ± 7	266 ± 7
	3019	Pl	1.96	52.83	92	174 ± 6	178 ± 6
CINTO CORSICA Ignimbrites (1st cycle)	1	Fs	5.07	71.53	93	93 ± 3	95.5± 3
	1 nm.	Fs	2.84	58.27	88	134 ± 4	137 ± 4
	1 m.	Fs	9.59	110.4	94	76.5± 2	78 ± 2
	2	Fs	2.19	85.80	93	248 ± 7	253 ± 7
	2	Am	2.24	127.8	97	351 ± 11	358 ± 11
	3	Fs	1.66	212.8	97	713 ± 18	724 ± 18
	4	Fs	3.68	110.2	93	192 ± 5	197 ± 5
	4	Cl	3.37	103.2	86	197 ± 6	201 ± 6
	5	Fs	4.83	149.5	93	199 ± 6	203 ± 6
	5	Cl	3.25	84.02	91	167 ± 5	171 ± 6
	6	Fs	4.13	67.2	79	107 ± 3	110 ± 3
	6	Cl	2.22	48.32	74	142 ± 5	145 ± 5
CORSICA Andesites (1st cycle)	3032	Fs	0.550	22.05	54	253 ± 9	259 ± 9
	3032	Cl	1.04	45.11	95	273 ± 8	279 ± 9
NURRA (SARDINIA) Ignimbrites	104	Bt	6.05	287.1	93	297 ± 9	303 ± 9
	149	Fs	6.67	256.9	95	244 ± 9	249 ± 9
	149	Bt	8.31	393.5	95	296 ± 8	302 ± 8

TABLE II (*continued*)

Sample			K_2O (%)	rad.^{40}Ar (10^{-11} mol/g	100.rad.^{40}Ar / Total^{40}Ar	Calculated age (± 1σ) in Ma (1) (a)	Calculated age (± 1σ) in Ma (2) (b)
Location and Rock type	Sample Number	Mineral					
GALLURA (SARDINIA) (1st cycle ?) Ignimbrites	1	Whole rock	4.90	186.8	97	242 ± 7	247 ± 7
	2 100-160μm	Fs	4.86	156.8	96	207 ± 6	211 ± 6
	2 160-250μm	Fs	6.50	224.8	87	221 ± 6	226 ± 6
	3	Fs+Q	3.84	153.1	97	252 ± 7	258 ± 7
	4	Fs+Q	5.60	232.3	95	262 ± 7	267 ± 7
	4	Q+Fs	0.200	8.116	76	256 ± 9	262 ± 9
	5	Fs+Q	4.60	177.2	90	244 ± 7	249 ± 7
	5	Fs+Q	4.39	176.7	96	254 ± 9	260 ± 9
ESCALAPLANO (SARDINIA) Andesite	76	Pl	0.560	23.06	89	260 ± 8	266 ± 8
	76	Bt	7.39	349.1	87	295 ± 8	301 ± 8
SEUI-SEULO (SARDINIA) Ignimbrites	88	Pl	1.06	43.52	83	259 ± 7	265 ± 7
	164	Fs	2.46	101.8	91	261 ± 8	267 ± 8
BAUNEI (SARDINIA) Ignimbrites	151	Pl	0.870	39.02	86	281 ± 10	287 ± 10
	153	Pl	0.600	27.91	69	291 ± 10	297 ± 11
	153	Cl+Bt	2.06	43.24	47	137 ± 5	140 ± 5
	154	Fs+Q	2.52	116.7	90	290 ± 9	296 ± 9
TALANA-VILLAGRANDE (SARDINIA) Ignimbrites	155 nm.	Fs+Q	1.43	59.43	87	262 ± 8	268 ± 8
	155 m.	Fs+Q	1.81	74.02	93	258 ± 8	264 ± 8
	155	Bt+Cl	2.59	125	84	301 ± 8	307 ± 8
	158	Pl	0.375	16.82	61	281 ± 10	287 ± 10
	159	Pl	0.405	20.06	61	308 ± 11	315 ± 11
Mt FERRU (SARDINIA) Ignimbrites	160	Pl	2.23	73.44	86	211 ± 8	215 ± 8
	161	Fs+Q	2.96	118.2	92	252 ± 8	258 ± 8

* not studied in paleomagnetism

(a) $\lambda_\epsilon = 0.585 \cdot 10^{-10}$/year
$\lambda_\beta = 4.72 \cdot 10^{-10}$/year
$^{40}K/^{tot}K = 1.19 \cdot 10^{-4}$ mol/mol.

(b) $\lambda_\epsilon = 0.581 \cdot 10^{-10}$/year
$\lambda_\beta = 4.962 \cdot 10^{-10}$/year
$^{40}K/^{tot}K = 1.167 \cdot 10^{-4}$ mol/mol.

Pl = plagioclases; *Bt* = biotites; *Hb* = hornblendes; *Fs* = undifferentiated feldspars; *Cl* = chlorites; *Q* = quartz; *m* = magnetic fraction (relative); *nm* = non-magnetic fraction.

three main volcanic complexes.

The directions of sites 149 and 104 in Nurra are very similar. The mean direction $D = 126°$, $I = -1°$ and the pole position $\lambda = 253°$ E, $\phi = 27°$ N differ from those yielded by southern-eastern samples. It has to be noted that similar directions have been obtained on the so called Permo-Triassic sandstones lying in the neighbourhood of the volcanics (Zijderveld et al. 1970; Edel, 1980).

K-Ar DATING

Analytical method

All the K-Ar determinations were performed on separated minerals. The 100—160 μm sieve fraction of the ground rock samples, was selected for magnetic separation of micas, amphiboles, chlorites and feldspars. Whenever necessary, further refinement was done with heavy liquids. Conventional argon determinations were made by isotope dilution and mass spectrometry (MS 10S). The technique has been more extensively described previously (Westphal et al., 1979). Potassium was measured by flame photometry using the lithium metaborate fusion technique (Besnus and Rouault, 1978). The results are listed in Table II. Dates were calculated with both sets of decay constants. Throughout the discussion, ages calculated with the former set are used since the geochronological time scale has not yet been modified to take into account the new determinations of the potassium decay constants. Plus-minus figures were estimates for analytical precision at one standard deviation. They were calculated following the procedure given by Cox and Dalrymple (1967).

Results and discussion

Corsica

Ota gabbro-diorite (second alkaline cycle). The five hornblende ages are moderately scattered while the coexisting plagioclases indicate systematically lower dates. The excellent agreement of the results yielded by hornblende and biotite 3017 allows us to evaluate the crystallization age with confidence around 280 Ma. Both hornblende and biotite are effectively argon-retentive and fairly insensitive to subsequent tectonic events (Maluski, 1977). Furthermore, the hornblendes show a slight but variable degree of chloritisation which can explain the moderate scattering of their ages.

Cinto-Scandola complex. The eruption of those volcanics took place between calc-alkaline granitic bodies and hyperalkaline granites which are contemporaneous of the gabbro-diorite. ^{39}Ar-^{40}Ar determinations (Maluski, 1977) attributed an age of 330 Ma to the calc-alkaline granites. One could, therefore, safely expect for the Cinto-Scandola complex dates of between 280 and 330 Ma. None of them falls in the expected range. Two values are

higher than 330 Ma and these probably correspond to occurrences of excess argon. The remaining ages are lower than 280 Ma. The investigated systems, either feldspars or chlorites, are particularly sensitive to tectonic events and have probably been disturbed by the intense Alpine tectonics. Maluski (1977), for instance, investigated the influence of tectonics on the ^{39}Ar-^{40}Ar ages of granites from the same region. Whenever the granites had been deeply affected by Alpine tectonics, ^{39}Ar-^{40}Ar data on feldspars and, most particularly, on plagioclases were geologically meaningless. Our results, though disappointing, are therefore consistent with the conclusion of this previous investigation.

Sardinia

Gallura. All the dates but one fall within the 240-260 My range. The meaning of ages is difficult to evaluate since the K-Ar determinations have been performed on alkali feldspars. These systems are reliable K-Ar clocks only in areas unaffected by tectonics (Albarede et al., 1978). Since the volcanic complexes of Gallura have been moderately faulted, the dates are likely to represent minimum ages of crystallization.

Nurra. The biotites indicate identical ages which lie at the Stephanian-Westphalian boundary. One feldspar yields a lower date. Since biotites constitute highly-reliable K-Ar clocks, one can safely interpret their K-Ar ages as those of crystallization.

Southeast. One biotite, one chlorite and several feldspars indicate Stephanian or Upper Westphalian ages. The agreement among dates yielded by different K-Ar clocks lead us to assess the age of that ignimbritic volcanism between 280 and 310 Ma. Several arguments allow us to reject the lower values, particularly those between 250 and 260 Ma.

The coexisting biotite of plagioclase 76 yields an age of 295 Ma. Furthermore, this feldspar shows evidence of tectonic stress.

In the light of the Tertiary ignimbritic cycle (Montigny et al., 1981) it seems unlikely that the Paleozoic one lasted over 70 Ma.

All of those low age values are yielded by chlorite or feldspars which might have been slightly disturbed by Alpine tectonics.

Conclusion

Our K-Ar determinations allow us to establish the age of the second volcanic cycle of Corsica as approximately 280 Ma. This result is in agreement with those obtained by ^{39}Ar-^{40}Ar (Maluski, 1977) and U-Pb (Lancelot in Maluski, 1977) on hyperalkaline granites of the same area. The first cycle, because of a lack of proper K-Ar clocks, i.e. amphiboles or biotites, cannot be determined. Nevertheless, the ^{39}Ar-^{40}Ar determinations on granites and the field observations lead us to assign the event within the 280—330 Ma range.

In Sardinia, the K-Ar dates on biotites and some feldspars, indicate Stephanian—Upper Westphalian ages for the volcanics of Nurra and the

southeast. On the other hand, an uncertainty remains in Gallura where alkali feldspars exhibit Permian ages. The nature of the K-Ar clock, however, suggests in that case a possible overprinting of alpine tectonics. It is, therefore, likely that all of the Paleozoic volcanics of Sardinia are contemporaneous, i.e. lower Stephanian or Upper Westphalian in age. A close examination of the paleontological record, as mentioned already, does not contradict such a statement.

DISCUSSION

Upper Paleozoic calc-alkaline volcanics occur in Corsica, Sardinia, northern Italy (Bolzano) (Vellutini, 1977) and in Spain (Atienza) (Hernando et al., 1981). In Maures, the rhyolites interbedded with Stephanian sediments may belong to this volcanic phase (Zijderveld, 1975). The alkaline cycle is found in Esterel, Corsica, Sardinia and northern Italy (Vellutini, 1977). On the other hand, Calabiar may have been in contact with Sardinia until the Upper Tertiary (Alvarez et al., 1974; Wezel, 1974). The Paleozoic nappes of Sila show structures similar to those of the southeastern Sardinian basement:

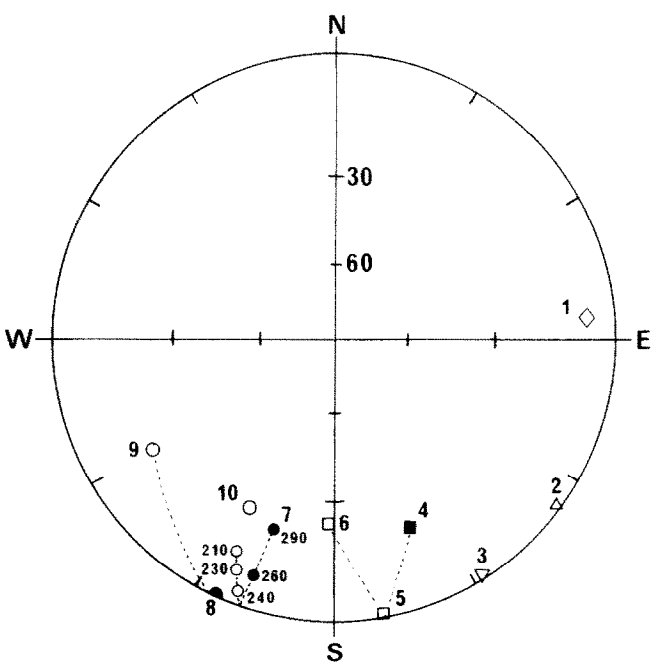


Fig. 7. Stereogram of the mean directions obtained on Upper Paleozoic volcanics of southeast Sardinia (*1*) (our study), northwest Sardinia (*2*) (our study), North Sardinia (Gallura) (*3*) (Zijderveld et al., 1970; Westphal et al., 1976; Storetvedt and Markhüs, 1978), Corsica (*4*, *5*: first cycle; *6*: second cycle) (Westphal, 1976; completed), Esterel (*9*: Stephanian rhyolites; *8*: Stephanian sediments; *10*: second cycle) (Zijderveld, 1975). Directions of Eurasia from Irving's (1977) poles (*7*, 290—210 Ma).

TABLE III

Paleomagnetic and geochronologic data on the Upper Paleozoic outcrops of Maures, Esterel, Corsica, Sardinia, Calabria (Sila)

	Formations	Ages (Ma)	References	D°	I°	α_{95}	References
Esterel (S.E. France)	rhyolites	290 ± 10	Roubault et al., 1970	239	-16	10.7	Zijderveld, 1975
	arkoses	Stephanian	Zijderveld, 1975	205	0.5	12.2	Zijderveld, 1975
	rhyolites and basalts (2nd cycle)	270 ± 10 to 228 ± 10	Roubault et al., 1970	207	-23.5	6.1	Zijderveld, 1975
	sediments	Permian	Zijderveld, 1975	206	-17.5	6.5	Zijderveld, 1975
Scandola-Cinto (Corsica)	andesites (1st cycle)	≥273 ± 8	This work	157	19	35	Westphal, 1976 and unpublished.
	ignimbrites (1st cycle)	>280 ?		170	- 1	19	"
	ignimbrites, gabbros , dolerites (2nd cycle)	283 ± 10	This work	181	-24	9	Westphal, 1976 and unpublished
Nurra (N.W. Sard.)	ignimbrites	296 ± 9	This work	126	- 1	-	This work
Gallura (N. Sard.)	ignimbrites (1st cycle)	≥255	This work	147	- 1	7	Zijderveld et al., 1970 Westphal et al., 1976 Storetvedt and Markhüs, 1978.
Barbagia Ogliastra Gerrei (S.E. Sard.)	ignimbrites and andesites (1st cycle ?)	(310 ?) 300-280	This work	85	- 7	5.5	This work
Sila (central Calabria)	diorites, amphibolites, dykes	260 ± 10	Borsi and Dubois, 1968	57	- 7	9.5	Manzoni, 1979

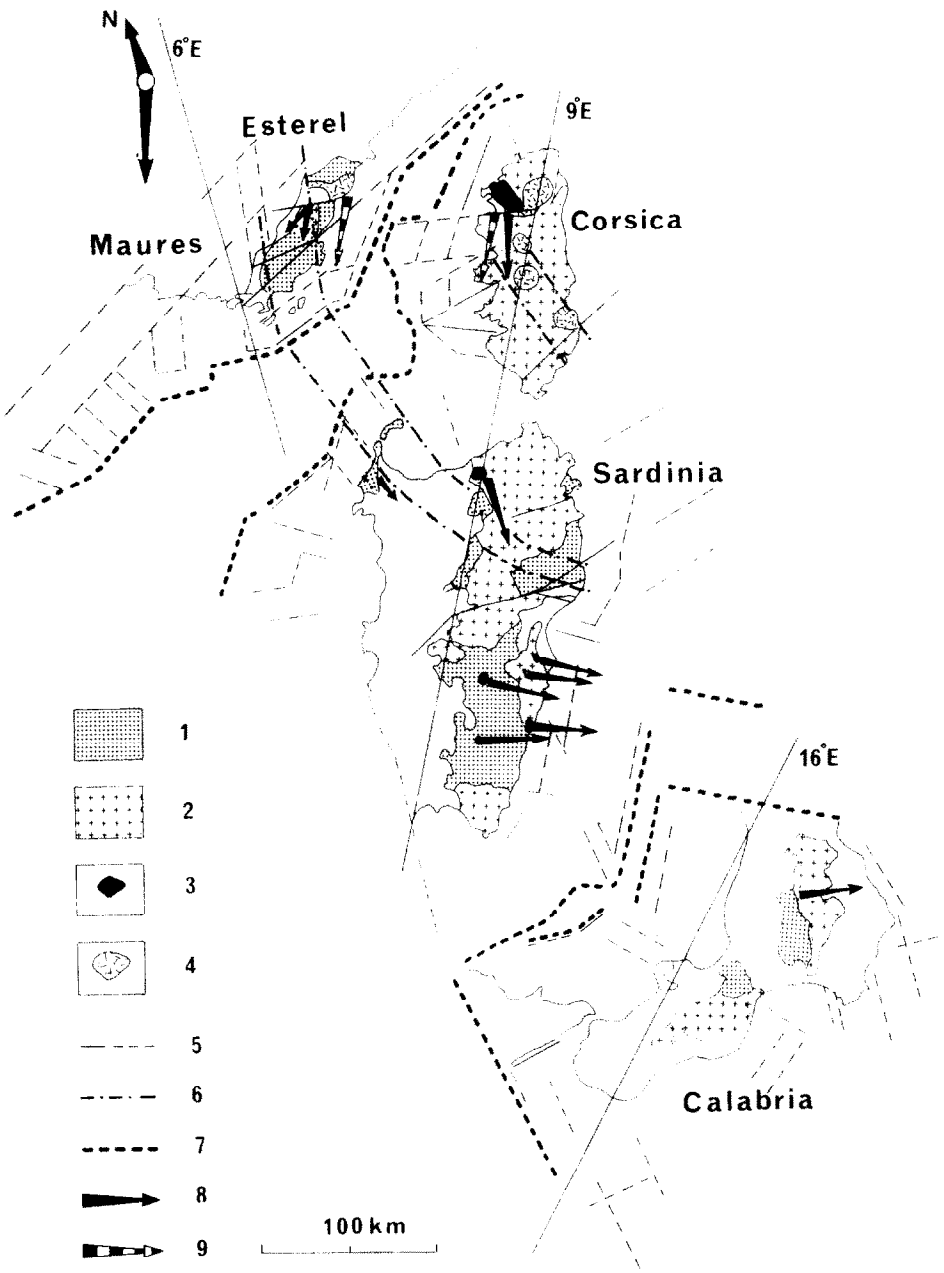


Fig. 8. Possible reconstruction of Maures—Esterel—Corsica—Sardinia—(Calabria?) in Lower Permian time. *1* = Paleozoic; *2* = granites and granodiorites; *3* = Westphalian—Stephanian calc-alkaline volcanics; *4* = Upper Stephanian alkaline volcanics; *5* = basement discontinuities from gravity, magnetism and bathymetry; *6* = metamorphic and magmatic zonations; *7* = possible continental limits; paleomagnetic directions from *8* (=the 1st calc-alkaline cycle); *9* = the 2nd alkaline cycle.

acidic to basic dikes and granites intrude an ancient metamorphic basement in both cases. A K-Ar investigation on one biotite from a granite yields an age of about 260 Ma (Borsi and Dubois, 1968). Since this determination is unique, we regard the question of the granite ages as still unsolved. The similar nature of the materials, therefore, leads us to compare the paleomagnetic results bearing on both regions.

The results, indicated in Table III and illustrated by Fig. 7, display a great heterogeneity in the paleomagnetic directions with the geographical location of the Paleozoic outcrops. Since all mean directions are likely to be representative of the magnetic field when the volcanism has taken place, the diverging directions have to be interpreted in terms of block rotations in Late Hercynian times.

The second volcanic cycle which has taken place in Esterel, Corsica, Sardinia and Italy, corresponds to an extension phase, featured by emplacement of basalts and gabbros (Vellutini, 1977). Paleomagnetic studies on these volcanics are only available in Esterel and in northwest Corsica. If one assumes that the Corso-Sardinian block has remained undeformed subsequent to the Stephanian—Autunian boundary, a reconstruction based on parallelism of directions obtained in Corsica and Esterel can be proposed for the Lower Permian (Fig. 8).

The first volcanic cycle of Corsica, Sardinia and northern Italy, consists of andesites and ignimbrites (Vellutini, 1977). It is interesting to compare the Upper Paleozoic volcanism with the Oligo-Miocene calc-alkaline cycle of Sardinia and southern France. This latter effusive phase is connected with the opening of the Provence basin, the movements of Sardinia and a possible subduction zone dipping towards the north or northwest under Sardinia (Bellon et al., 1977). The calc-alkaline cycle and probably the Sardinian drift, lasted about 15 Ma. The 30° rotation which occurred at the end of the volcanic episode, was shortlived, 2 Ma at most (Edel, 1980; Montigny et al., 1981). Either in the Upper Paleozoic or in the Tertiary, the general tectonic context is a collision between the Eurasian and African plates. If the Upper Paleozoic volcanism has lasted as long as the Tertiary, the analytical uncertainties on the ages, i.e. ±10 Ma, does not allow us to establish whether the volcanics of Esterel, northwest Corsica, Nurra, southeast Sardinia and Gallura are contemporaneous or whether they succeed one another. Consequently the study of block motions in the Paleozoic becomes far more difficult than in the Tertiary. Paleomagnetic and radiometric data do not tell whether the different directions measured in Corsica and Sardinia represent different stages of a definite motion or differential motions.

Assuming that the calc-alkaline ignimbrites of Corsica and Gallura were formed at the same time, differences in declination imply an uncoupling of 20—25° between those volcanic areas. The prolongation of dikes from Gallura to southern Corsica (Arthaud and Matte, 1977) and the linearity of the granitic zonation in southern and middle Corsica (Orsini, 1976) suggest that the uncoupling has occurred between middle and northern Corsica. In the

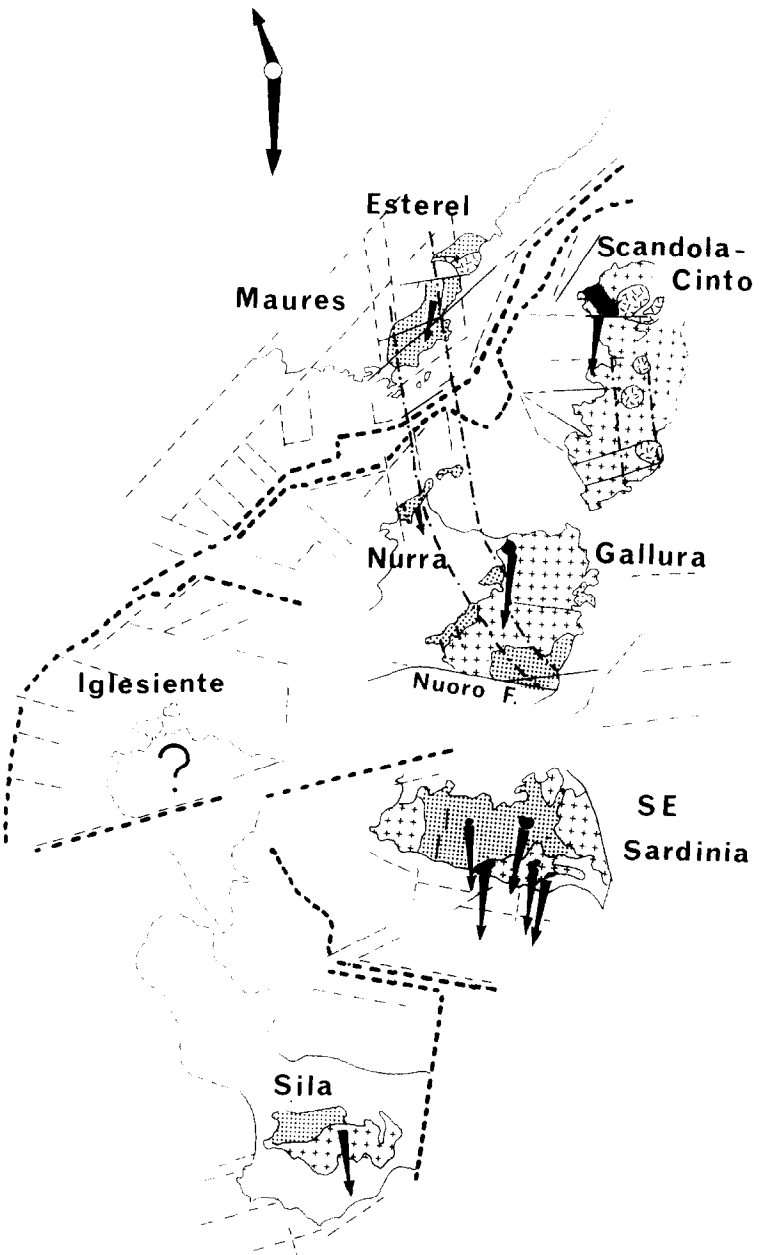


Fig. 9. Possible reconstruction of Maures—Esterel—Corsica—Sardinia—(Calabria?) in Westphalian—Stephanian time.

same way, contemporaneous volcanism in Gallura and southeast Sardinia leads us to deduce an uncoupling of 60° between those areas. The anticlockwise rotation might have been caused by a curvature of the basement or a strike slip motion along the Nuoro-fault. Nevertheless, the existence of eastern declinations similar to those of southeast Sardinia in the lower andesitic layers of northwest Corsica (Ashworth and Nairn, 1965; Westphal, 1967) would refute uncoupling of the different areas of the Corso-Sardinian plate.

The proposed reconstruction is only tentative (Fig. 9). It is based on parallelism of paleomagnetic directions yielded by the Stephanian sediments of Esterel, similar to those of stable Europe (Irving, 1977), the ignimbrites of the first cycle in Corsica and the ignimbrites of the first cycle in Sardinia. The fit of different blocks is made possible by the continuity of metamorphic zonations from Maures, to eastern Sardinia through Nurra (Westphal et al., 1976). Northwest Corsica is then uncoupled from the rest of Corsica and Gallura. Iglesiente (southwest Sardinia) probably lies in a different position since the basement does not show the metamorphic zonation mentioned above. The position attributed to Calabria is highly speculative since the paleomagnetic data have been obtained on materials involved in nappes.

CONCLUSION

The K-Ar investigation combined with field data indicates a Carboniferous age for most of the Upper Paleozoic volcanics of Sardinia and Corsica. An uncertainty, however, remains for the ignimbritic series of Gallura.

The paleomagnetic directions obtained in the three areas of southeast Sardinia (Barbagia—Ogliastra—Gerrei) and in the northwest (Nurra) are different from those determined by other authors in Gallura and in northwest Corsica. The difference between southeast Sardinia and northwest Corsica on volcanics of probably identical age amounts to 85°, while inclinations are similar. The variability of mean declinations with geographical areas suggests the existence of an important phase of deformation in Late Hercynian times provided the basement has not subsequently been affected by intense tectonics (Arthaud and Matte, 1977).

The calc-alkaline volcanism may be related to a subduction zone and consequently to a plate collision. In Late Hercynian times the collision between Eurasia and Gondwana still continues (Irving, 1977). According to the age determinations on calc-alkaline volcanics these events would have occurred during the Asturian phase. The nature of the movement consists either of an anticlockwise rotation of about 90° of the whole Corso-Sardinian plate, or differential rotations, the most important happening between north and southwest Sardinia.

ACKNOWLEDGEMENTS

We thank Prof. W. Lowrie for fruitful discussions of our paleomagnetic data. Our collegue M. Westphal kindly reviewed the manuscript. The work

was supported by E.R.A. 683 from C.N.R.S. and A.T.P. "Geodynamique" from I.N.A.G.

REFERENCES

Albarede, F., Feraud, G., Kaneoka, I., Allegre, C.J., 1978. $^{39}A-^{40}A$ dating: the importance of K-feldspars on multi-mineral data of polyorogenic areas. J. Geol., 86: 581–598.

Alvarez, W., Cocozza, T. and Wezel, F.C., 1974. Fragmentation of the Alpine orogenic belt by microplate dispersal. Nature, 248: 309–314.

Arthaud, F. and Matte, P., 1977. Détermination de la position initiale de la Corse et de la Sardaigne à la fin de l'orogénèse hercynienne grâce aux marqueurs géologiques antémésozoïques. Bull. Soc. Géol. Fr., 7, 19 (4): 833–840.

Ashworth, T.P. and Nairn, A.E.M., 1965. An anomalous pole from Corsica. Palaeogeogr., Palaeoclimatol., Palaeoecol., 1: 119–125.

Bellon, H., Coulon, C. and Edel, J.B., 1977. Le déplacement de la Sardaigne. Synthèse des données géochronologiques, magmatiques et paléomagnétiques. Bull. Soc. Géol. Fr., 7, 19(4): 825–832.

Besnus, Y. and Rouault, R., 1978. Le point sur les méthodes d'analyse des roches utilisées en laboratoire de spectrométrie. Note technique, Institut de Géologie de Strasbourg.

Bonin, B., Vialette, Y. and Lameyre, J., 1972. Géochronologie et signification du complexe granitique annulaire de Tolla-Cauro. C.R. Soc. Géol. Fr., 9: 52–57.

Borsi, S. and Dubois, R., 1968. Données géochronoloques sur l'histoire hercynienne et alpine de la Calabre centrale. C.R. Acad. Sci. Paris, 266: 72–75.

Cox, A. and Dalrymple, G.B., 1967. Statistical analysis of geomagnetical reversal data and the precision of potassium argon dating. J. Geophys. Res., 72: 2603.

Cozzupoli, D., Dicendenti, A., Lombardi, G. and Nicoletti, M., 1971. Chronologia K-Ar delle manifestazioni eruttive dell settore Seui-Seulo (Barbagia, Sardegna). Period. Mineral., 40 (1–2): 113–124.

Edel, J.B., Etude paléomagnétique en Sardaigne. Conséquences pour la géodynamique de la Méditerranée occidentale. Thesis, Strasbourg, 310 pp.

Hernando, S., Schott, J.J., Thuizat, R. and Montigny, R., 1980. Age des andésites et des sédiments interstratifiés de la région d'Atienza (Espagne): étude stratigraphique, géochronologique et paléomagnétique. Sci. Geol., Bull., 33(2): 119–128.

Irving, E., 1977. Drift of the major continental blocks since the Devonian. Nature, 270: 304–309.

Lauro, C., Negretti, G. and Sbarracani, M.L., 1963. Contributo alla conoscenza delle formazioni permiane di Seui (Barbagia, Sardegna). G. Geol. (Bologna), 31 (1): 281–309.

Lombardi, G., Cozzupoli, D. and Nicoletti, M., 1974. Notizie petrograficha e dati sulla cronologia K-Ar del vulcanismo tardo paleozoico sardo. Period. Mineral., 43 (2): 1–92.

Maluski, H., 1974. Mise en évidence par la méthode $^{40}K-^{40}A$ de l'âge permien de certains granites hyperalcalins de Corse. C.R. Somm. Soc. Géol. Fr., 4: 93–95.

Maluski, H., 1977. Application de la méthode $^{40}Ar-^{39}Ar$ aux minéraux des roches cristallines perturbées par les évènements tectoniques et thermiques en Corse. Bull. Soc. Geol. Fr, 7, 19(4): 849–856.

Manzoni, M., 1979. Paleomagnetic evidence for non-apenninic origin of the Sila nappes (Calabria). Tectonophysics, 60: 169–188.

Montigny, R., Edel, J.B. and Thuizat, R., 1981. The rotation of Sardinia in the Tertiary: K-Ar dating and paleomagnetism. Earth Planet Sci. Lett., 54(2).

Nairn, A.E.M. and Westphal, M., 1968. Possible implication of the paleomagnetic study of Late Paleozoic igneous rocks of northwestern Corsica. Paleogeogr., Paleoclimatol., Paleoecol., 5: 179–204.

Novarese, V., 1917. L'Autuniano in Sardegna. Boll. Soc. Geol. Ital., 3b: 88—90.
Orsini, J., 1976. Les granitoïdes hercyniens corso-sardes: mise en évidence de deux associations magmatiques. Bull. Soc. Géol. Fr., 7, 18 (5): 1203—1206.
Pecorini, G., 1962. Nuovi osservazioni sul Permico della Nurra (Sardegna nord-occidentale). Accad. Naz. dei Lincei., Roma, 3, Ser. VIII, XXXII: 377—380.
Pecorini, G., 1974. Nuovi osservazioni sul Permo-trias di Escalaplano (Sardegna sud-orientale). Boll. Soc. Geol. Ital., 93: 991—999.
Roubault, M., Bordet, P., Leutwein, F., Sonet, E. and Zimmermann, J.L., 1970. Ages absolus des formations volcaniques du Tanneron-Estérel Var). C.R. Acad. Sci., 271: 1157—1160.
Storetvedt, K.M. and Markhüs, L.A., 1978. Multivectorial magnetization in Late Paleozoic volcanics from North-Sardinia; partial remagnetization and rotation. Geophys. J.R. Astron. Soc., 53: 245—257.
Storetvedt, K.M. and Petersen, N., 1976. Postulated rotation of Corsica not confirmed by new palaeomagnetic data. J. Geophys., 42: 59—71.
Vellutini, P., 1973. Etude géologique de la Vallée du Fango. Les Rhyolites ignimbritiques et leur Substratum paléozoïque. Thesis, Marseille, 214: 34 pp.
Vellutini, P., 1977. Le Magmatisme permien de la Corse du Nord-Ouest. Thesis, Aix-Marseille, 317 pp.
Westphal, M., 1967. Etude paléomagnétique de Formations volcaniques primaires de Corse. Rapports avec la Tectonique du Domaine ligurien. Thesis, Strasbourg, 52 pp.
Westphal, M., 1976. Paléomagnétisme des formations permiennes de Corse. Comparaison avec la Sardaigne et l'Estérel. Bull. Soc. Géol. Fr., 18(5): 1209—1215.
Westphal, M., Orsini, J. and Vellutini, P., 1976. Le micro-continent corso-sarde, sa position initiale: données paléomagnétiques et raccords géologiques. Tectonophysics, 30: 141—157.
Westphal, M., Montigny, R., Thuizat, R., Bardon, C., Bossert, A., Hamzeh, R. and Rolley, J.P., 1979. Paléomagnétisme et datation du volcanisme permien, triasique et crétacé du Maroc. Can. J. Earth. Sci., 16 (11): 2150—2164.
Wezel, F.C., 1974. Flysch successions and the tectonic evolution of Sicily during the Oligocene and early Miocene. In: C, Squires (Editor), Geology of Italy. Petrol. Explor. Soc. Lybia, pp. 1—23.
Zijderveld, J.D.A., 1975. Paleomagnetism of the Esterel Rocks. Thesis, Utrecht, 199 pp.
Zijderveld, J.D.A., De Jong, K.A. and Van der Voo, R., 1970. Rotation of Sardinia: paleomagnetic evidence from Permian rocks. Nature, 226: 933—934.